THE DUALIZING SHEAF RESTRICTS TO A RATIONAL TAIL WITH NEGATIVE DEGREE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-curves/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the dualizing short exact sequence, the identification of the tail with P^1 over the extension field k', the degrees -2[k' : k] and [k' : k], and the additive conclusion deg(omega_X|_C) = -[k' : k] are all retained verbatim.

Editorial note. The label lemma-rational-tail-negative names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Short exact sequence for the dualizing sheaf along a nodal closed subscheme. Same chapter, Lemmas lemma-closed-subscheme-reduced-gorenstein, lemma-facts-about-nodal-curves, lemma-closed-subscheme-nodal-curve. Gives the exact sequence 0 -> omega_C -> omega_X|_C -> O_{C cap X'} -> 0 on which the degree computation runs.
- The rational tail as a projective line over an extension field. Same chapter, Example example-rational-tail and Proposition proposition-projective-line. Identifies C with P^1_{k'}, so deg(omega_C) = -2[k' : k], and fixes the intersection degree deg(C cap X') = [k' : k].

Source reference. Original reference 0E3H: sources/stacks-curves/pinned-original-source.tex, source lines 5933–5933. Prerequisite pointer invoked at proof line(s) 5991; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0E38: sources/stacks-curves/pinned-original-source.tex, source lines 4794–4794. Prerequisite pointer invoked at proof line(s) 6002; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0E34: sources/stacks-curves/pinned-original-source.tex, source lines 1012–1012. Prerequisite pointer invoked at proof line(s) 6000; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original reference 0E37: sources/stacks-curves/pinned-original-source.tex, source lines 4763–4763. Prerequisite pointer invoked at proof line(s) 6001; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C6U: sources/stacks-curves/pinned-original-source.tex, source lines 2138–2138. Prerequisite pointer invoked at proof line(s) 6004; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be a field. Let X be a proper scheme over k having dimension 1 and $H^0(X, \mathcal{O}_X) = k$. Assume the singularities of X are at-worst-nodal. Let $C \subset X$ be a rational tail (Example 0E3H). Then* $\deg(\omega_X|_C) < 0$.

Proof. Let $X' \subset X$ be as in the example. Then we have a short exact sequence

$$0 \to \omega_C \to \omega_X|_C \to \mathcal{O}_{C \cap X'} \to 0$$

See Lemmas 0E34, 0E37, and 0E38. With k' as in the example we see that $\deg(\omega_C) = -2[k' : k]$ as $C \cong \mathbf{P}^1_{k'}$ by Proposition 0C6U and $\deg(C \cap X') = [k' : k]$. Hence $\deg(\omega_X|_C) = -[k' : k]$ which is negative. □